FINITE TOR DIMENSION FORCES INJECTIVITY ON THE CONORMAL LEVEL

THE STACKS PROJECT AUTHORS

Editorial note. No omission markers in-span; proof complete as written.

Allowed standard prerequisites (editorial).

- Obstruction lemma. lemma-not-finite-pd (same chapter, named). The contrapositive instance: the set-up data violate finite tor dimension.

Source reference. Original lemma 09PU: sources/stacks-dpa/pinned-original-source.tex, source lines 1755–1755. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $(A, \mathfrak{m})$ be a Noetherian local ring. Let $I \subset J \subset A$ be proper ideals. If A/J has finite tor dimension over A/I, then $I/\mathfrak{m}I \to J/\mathfrak{m}J$ is injective.*

Proof. Let $f \in I$ be an element mapping to a nonzero element of $I/\mathfrak{m}I$ which is mapped to zero in $J/\mathfrak{m}J$. We can choose an ideal I' with $\mathfrak{m}I \subset I' \subset I$ such that I/I' is generated by the image of f. Set $R = A/I$ and $R' = A/I'$. Let $J = (a_1, \ldots, a_m)$ for some $a_j \in A$. Then $f = \sum b_j a_j$ for some $b_j \in \mathfrak{m}$. Let $r_j, f_j \in R$ resp. $r'_j, f'_j \in R'$ be the image of b_j, a_j. Then we see we are in the situation of Lemma 09PU (with the ideal I of that lemma equal to $\mathfrak{m}_R$) and the lemma is proved. □

Date: stacks-project:dpa.tex:local-snapshot-20261012; GNU FDL 1.2 or later.